**Meron Ghirmai Tewelde**

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## Profile Summary

**Data-driven IT professional and developer with over four years of experience in software development, database architecture, and data infrastructure. Proven expertise in Python programming, SQL databases, and data processing pipelines, backed by strong quantitative foundations in statistics, probability, and linear algebra. Skilled in designing scalable systems, building search algorithms, and curating natural language corpora for under-resourced scripts. Passionate about applying AI/ML techniques and data science to solve regional challenges.**

## Professional Experience

**IT Specialist**

***Ministry of Land, Water & Environment – Asmara, Eritrea | 2020 – 2021***

- **Designed and deployed centralized relational databases (SQL/MS Access) that consolidated municipal records, land registry datasets, and environmental monitoring metrics for over 150 daily active personnel.**
- **Administered Windows Server infrastructure, implementing Active Directory protocols and network firewalls that achieved 99.8% server uptime.**
- **Built automated queries and reporting workflows to streamline routine data retrieval, significantly reducing manual report generation time.**

**Lead Developer & Software Engineer**

***Freelance | Jun 2022 – 2025***

- **Architected custom web applications, database backends, and user-facing digital systems for clients, ensuring data integrity and security.**

***Mekane Hiwet Android Application | 2025 – Present ( Scheduled Release: October 2026 )***

- **Engineered a digital library and text-processing application specifically for the Tigrinya-speaking community to address persistent local connectivity constraints and complex Ge'ez script rendering challenges.**
- **Designed the application with offline SQLite caching, real-time Firebase synchronization, and a custom image-rendering engine for excerpt sharing.**

## Technical Skills

- **Data Science & AI/ML: Python, Pandas, PyTorch, Data Engineering, Natural Language Processing (NLP), Data Modeling, Feature Extraction, SQL / Relational Databases**
- **Programming & Databases: Python, Java, C#, Node.js, SQL, MySQL, MS Access, Firebase (Firestore)**
- **Infrastructure & Tools: Git, Android SDK, Material Design 3, AWS, Azure, Docker, Windows Server, Active Directory, Firewalls**

## Education & Certifications

**Bachelor of Science in Computer Science ( GPA 3.82/4.00)**

***Eritrean Institute of Technology — Asmara, Eritrea | Graduated***

- **Quantitative Coursework: Statistics, Probability, Linear Algebra, Numerical Methods, Algorithm Design, Database Systems, Computer Networks, Software Engineering.**

**AIMS South Africa — AI/ML Program**

***African Institute for Mathematical Sciences — South Africa | Sept 2026***

- **Intensive track in Artificial Intelligence and Machine Learning foundations, complemented by ongoing Google DeepMind coursework in AI Research Foundations and Language Model Training.**

## Projects & Community Impact

- **Ge'ez NLP Corpus & Tokenizer: Sourced digitized historical manuscripts, educational texts, and community submissions to curate, clean, and annotate over 25,000 sentences (~350,000 tokens) to support NLP research for under-resourced languages. Open-sourced the initial dataset and custom tokenizer on GitHub.**
- **Eritrean Postal Service Capstone Project: Co-led a four-person team to develop a web-based database system and SMS notification pipeline digitizing legacy paper workflows across 14 regional branches. Designed the relational schema, optimized SQL queries, and implemented the backend API, streamlining package tracking for over 5,000 monthly transactions and reducing notification times from days to seconds.**

## Languages & Interests

- **Languages: Tigrinya (Native), English (Fluent), Amharic (Fluent), French (A2)**
- **Interests: Photography, Football, Reading**